

IT3091 – Machine Learning

Lecture 5

Supervised learning | model evaluation I: classification

Dr. Mahima Weerasinghe

Learning Aims

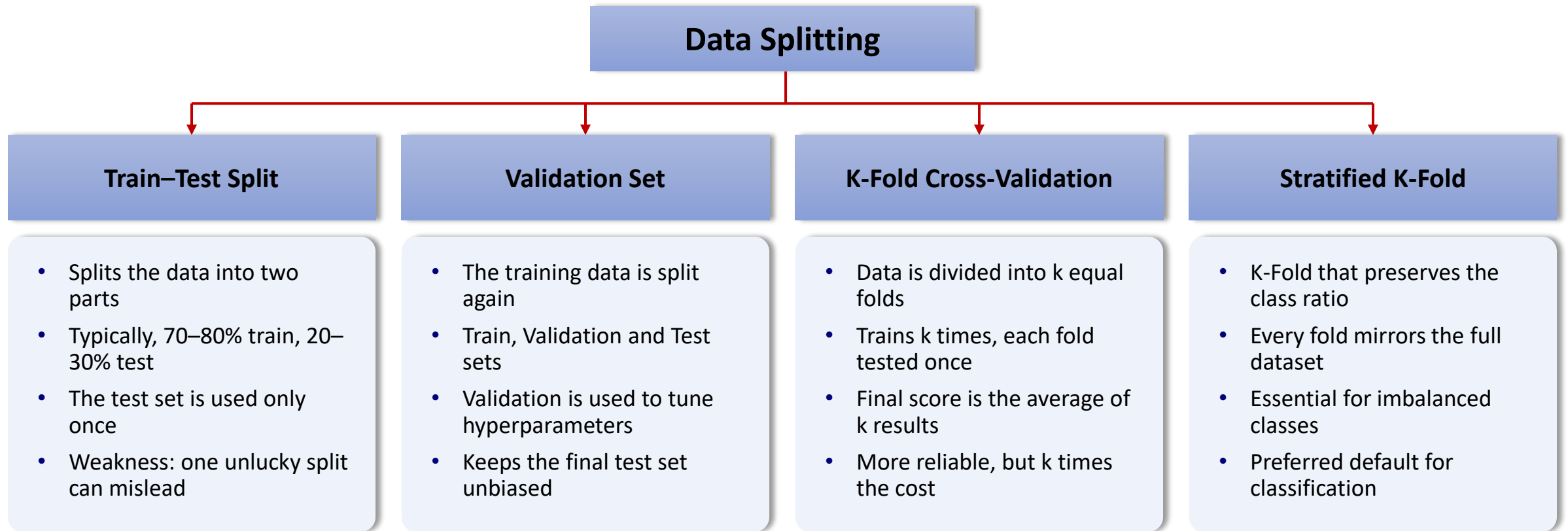
By the end of this lecture, students will be able to:

- ▶ Explain the concept and types of classification in supervised learning.
- ▶ Describe how Logistic Regression, kNN, Naïve Bayes, Decision Trees, and SVM perform classification.
- ▶ Interpret a confusion matrix using TP, TN, FP, and FN.
- ▶ Calculate and interpret Accuracy, Precision, Recall, and F1-score.
- ▶ Explain the purpose of ROC-AUC and Precision–Recall curves.
- ▶ Explain how decision thresholds affect false positives and false negatives.
- ▶ Distinguish between underfitting and overfitting using bias–variance intuition.
- ▶ Compare and select appropriate classification algorithms based on the problem and dataset characteristics



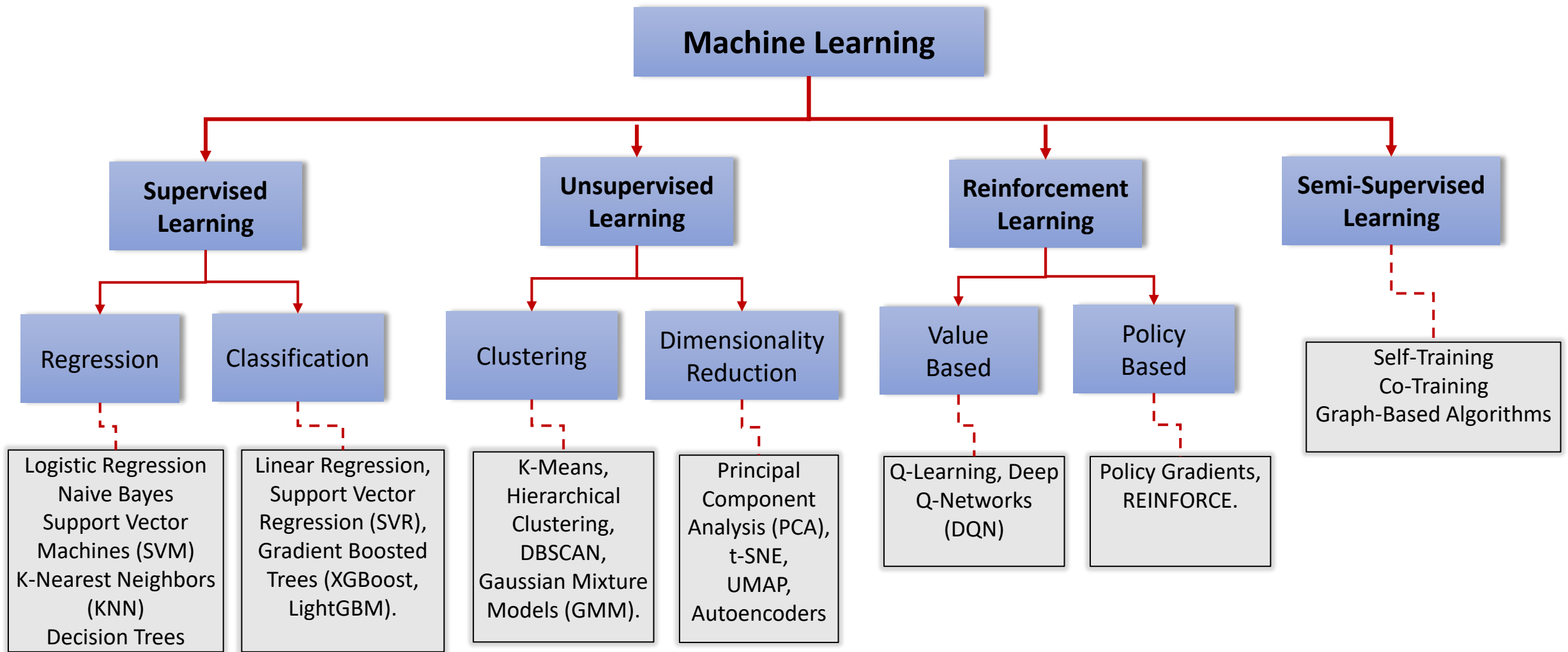
Data Splitting Techniques

- Data splitting separates data for training and evaluation so that a model can be tested on unseen data.



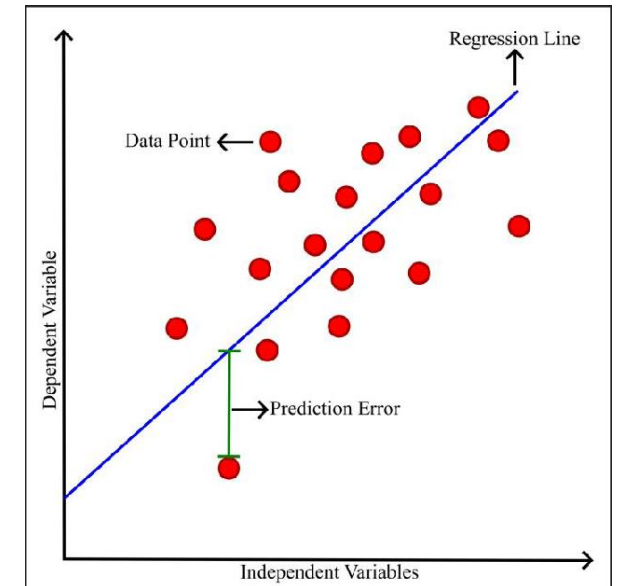
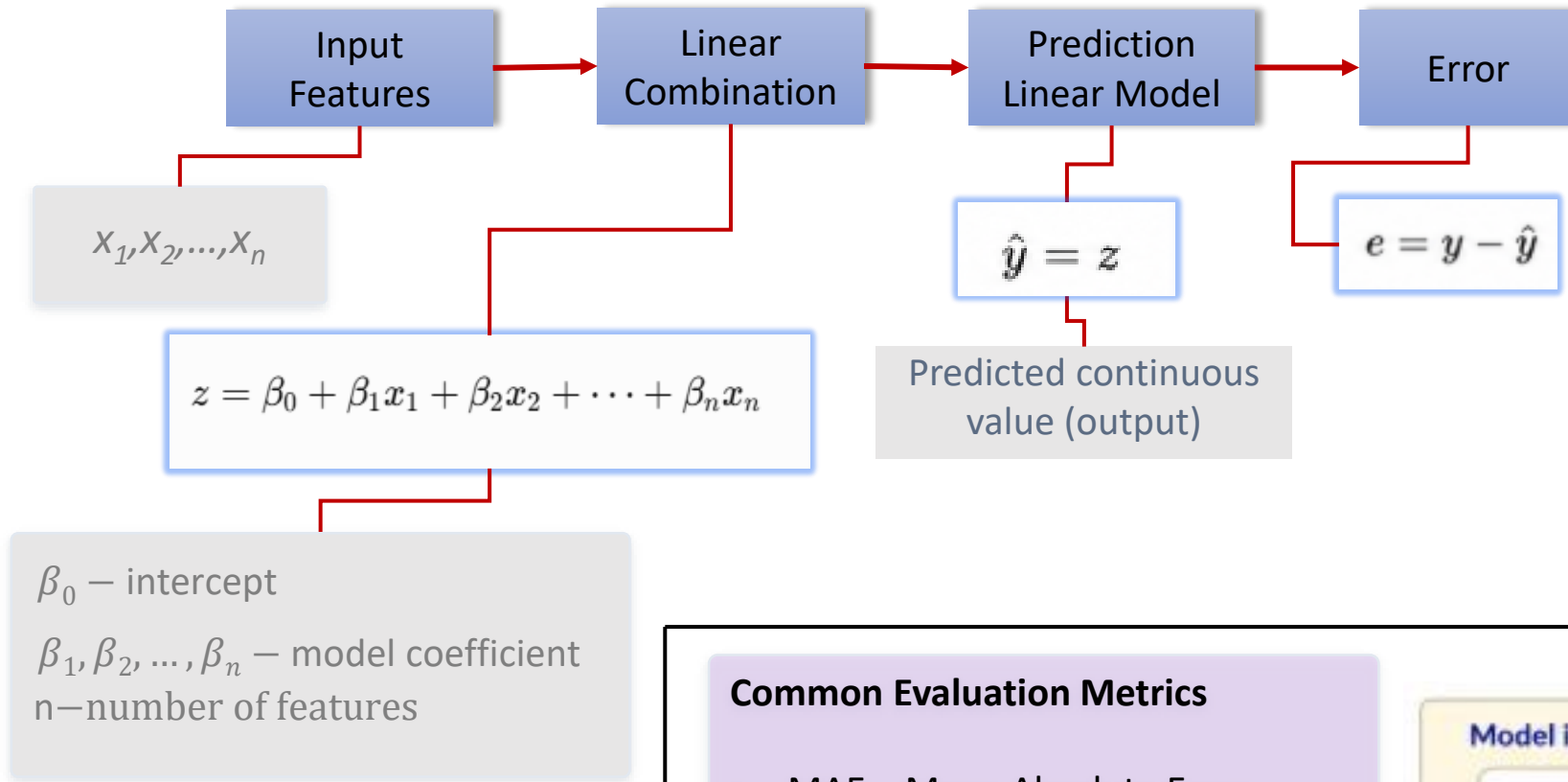
Key idea: cross-validation gives a more reliable estimate of performance than a single split, and stratification keeps each fold representative of the original class balance.

Machine Learning Techniques — Hierarchy



Linear Regression

- ▶ A supervised learning algorithm used to predict a continuous numerical value.



Common Evaluation Metrics

- MAE – Mean Absolute Error
- MSE – Mean Squared Error
- RMSE – Root Mean Squared Error
- R^2 – R Squared

Model is trained by minimizing the Sum of Squared Errors (SSE)

$$SSE = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

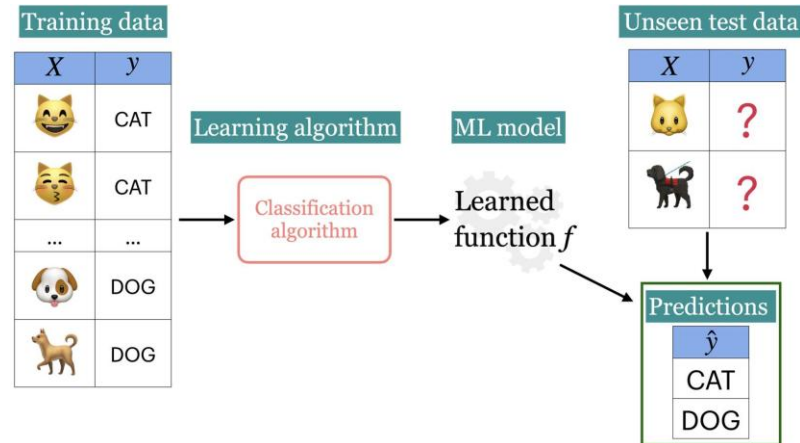
y_i – Actual value
 $\hat{y}_i$ – Predicted value
 m – Number of training samples

Introduction to Supervised Learning for Classification

- ▶ Classification is a supervised machine learning task that predicts predefined class labels using labeled training data.

Types of Features

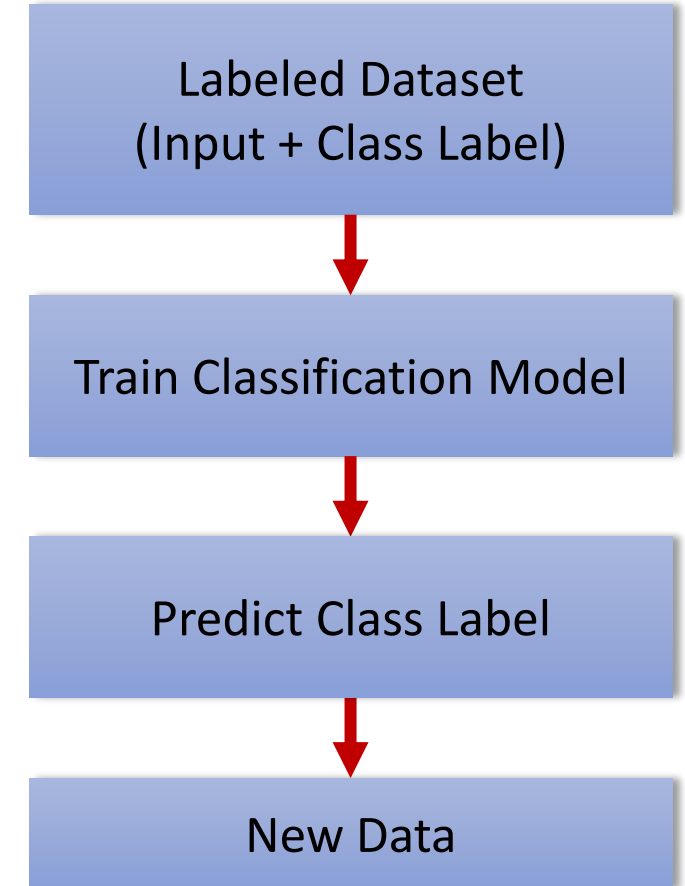
- Combining Features
- Mathematical Features
- Date & Time Features
- Domain-specific Features



Importance

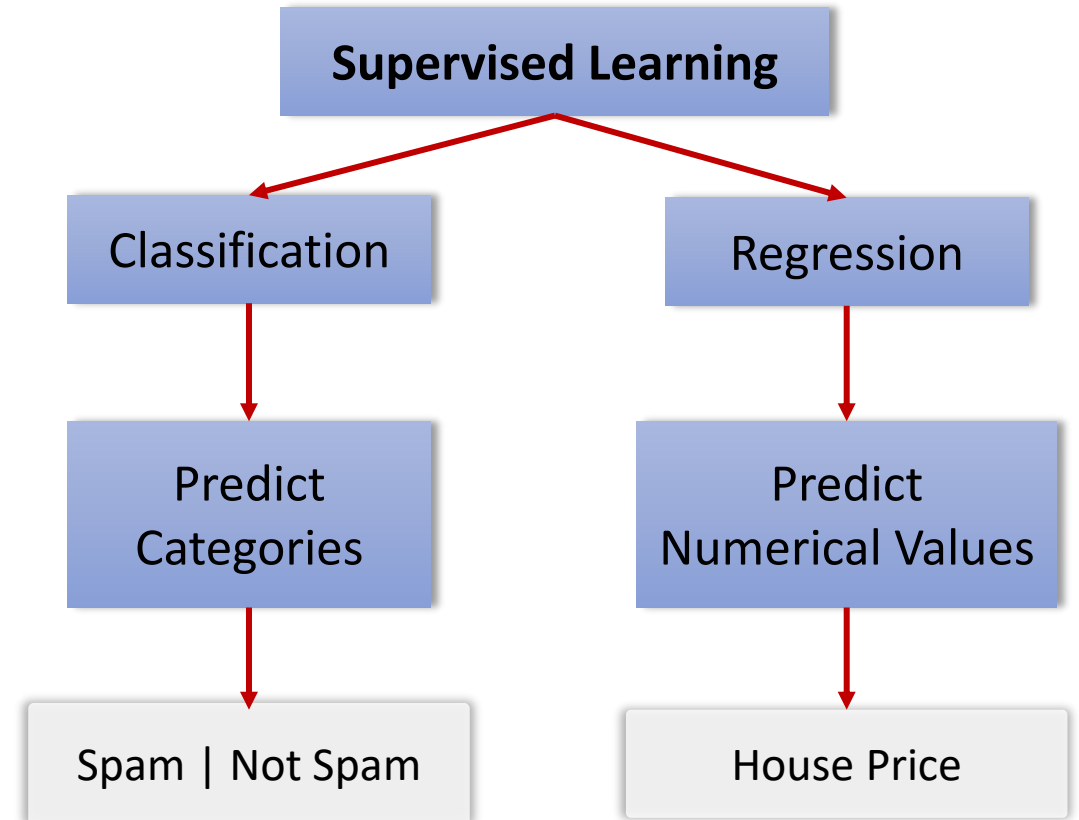
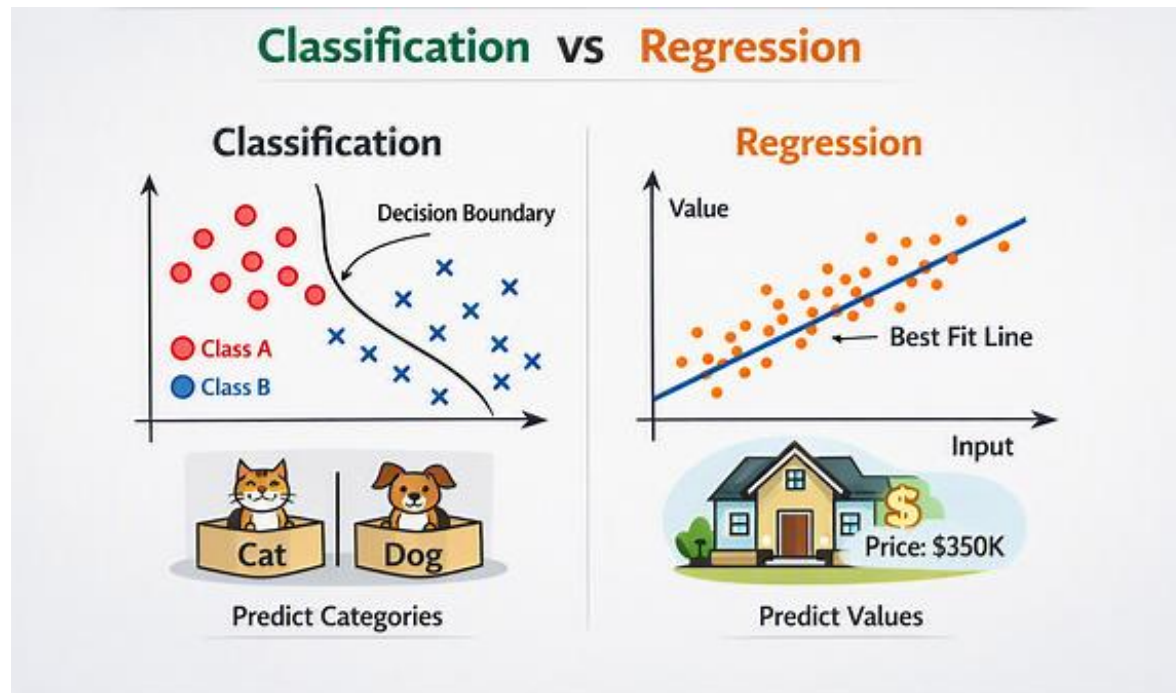
Classification enables machines to automatically categorize data and support intelligent decision-making.

Supervised Learning Workflow



Classification vs. Regression

- Classification predicts categories (class labels), while regression predicts continuous numerical values.



What is Classification?

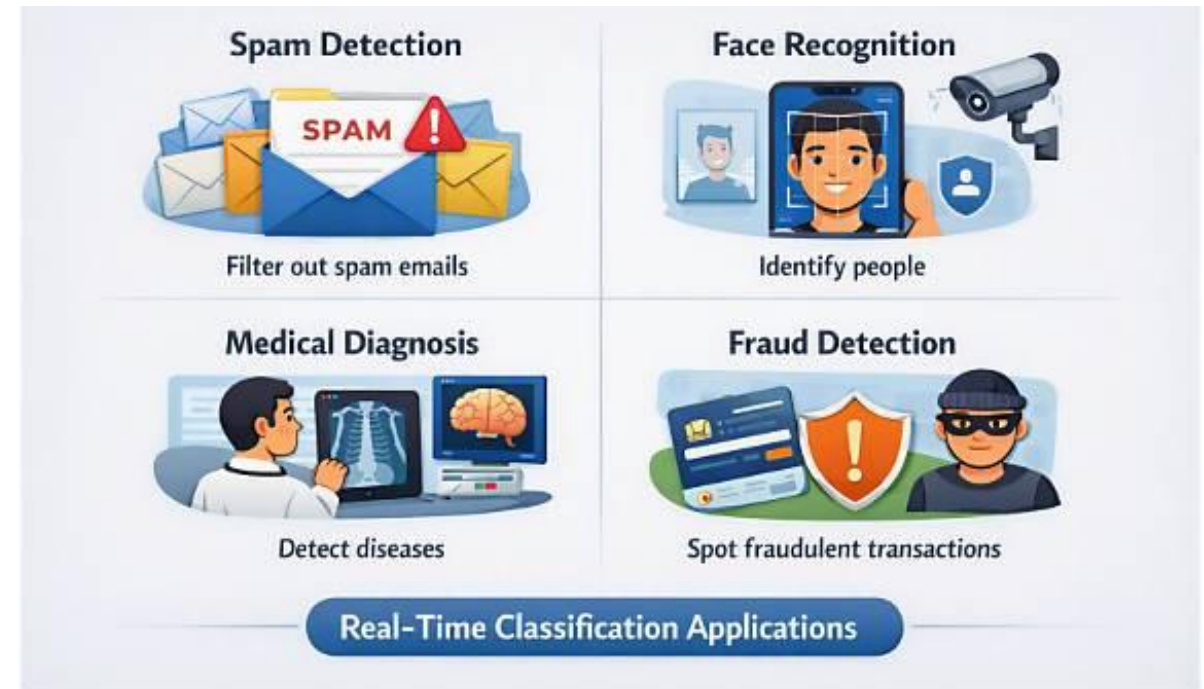
- Classification is a supervised learning task that assigns data into predefined categories (class labels) based on input features.

Characteristics

- Predicts class labels
- Learns from labeled data
- Used for decision making
- Output is categorical

Importance

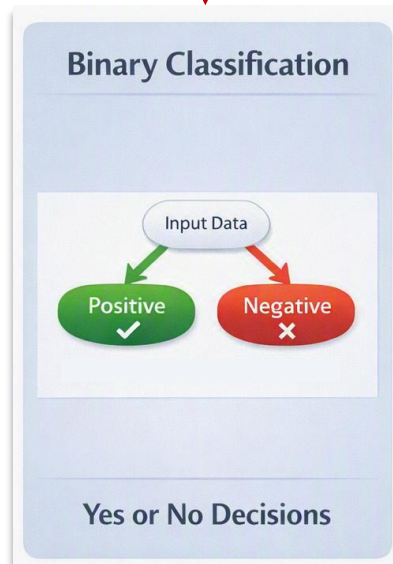
Classification enables intelligent systems to automatically categorize data and support decision-making across many real-world applications.



Types of Classification

Binary Classification

One input → One of Two Classes



Multiclass Classification

One input → One of Many Classes



Multilabel Classification

One input → Multiple Labels

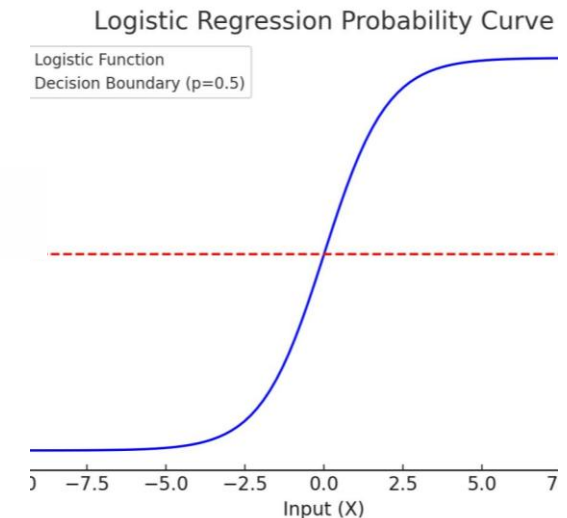
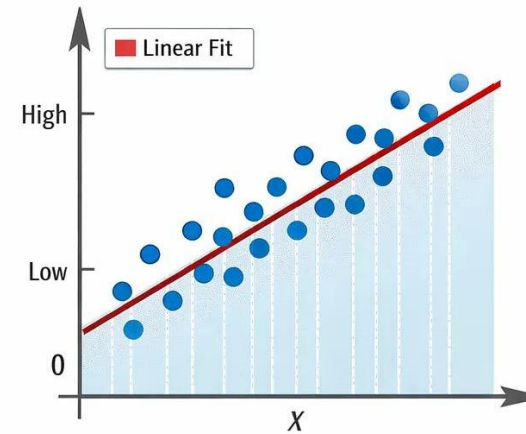


Why Linear Regression is Not Suitable for Classification

- ▶ Linear Regression predicts continuous values, whereas classification requires discrete class labels or probabilities.

Problems with Linear Regression

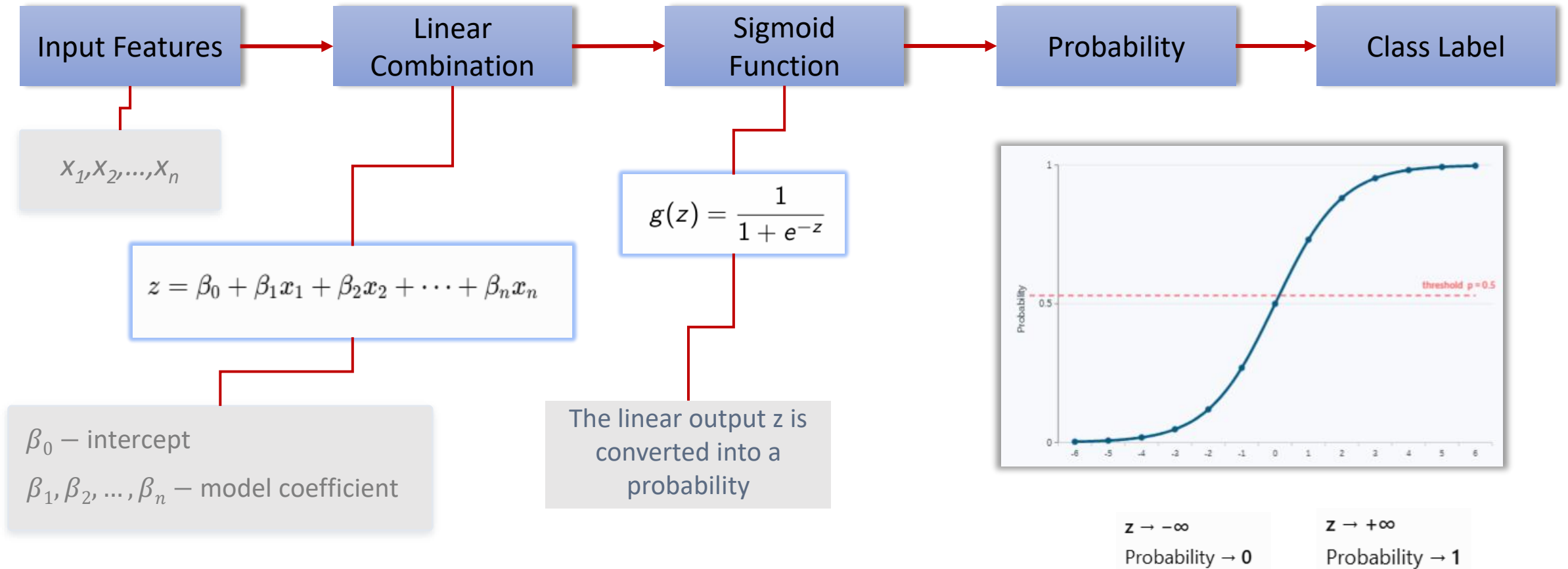
- Predicts continuous outputs
- Predictions can be less than 0 or greater than 1
- Cannot estimate class probabilities
- Poor decision boundaries



Linear Regression	Logistic Regression
Continuous Output	Probability (0–1)
Best for Regression	Best for Classification
No Decision Boundary	Decision Threshold (0.5)

Logistic Regression

- ▶ Logistic Regression is a supervised learning algorithm used to model the probability of a categorical outcome, particularly for binary classification.



Logistic Regression

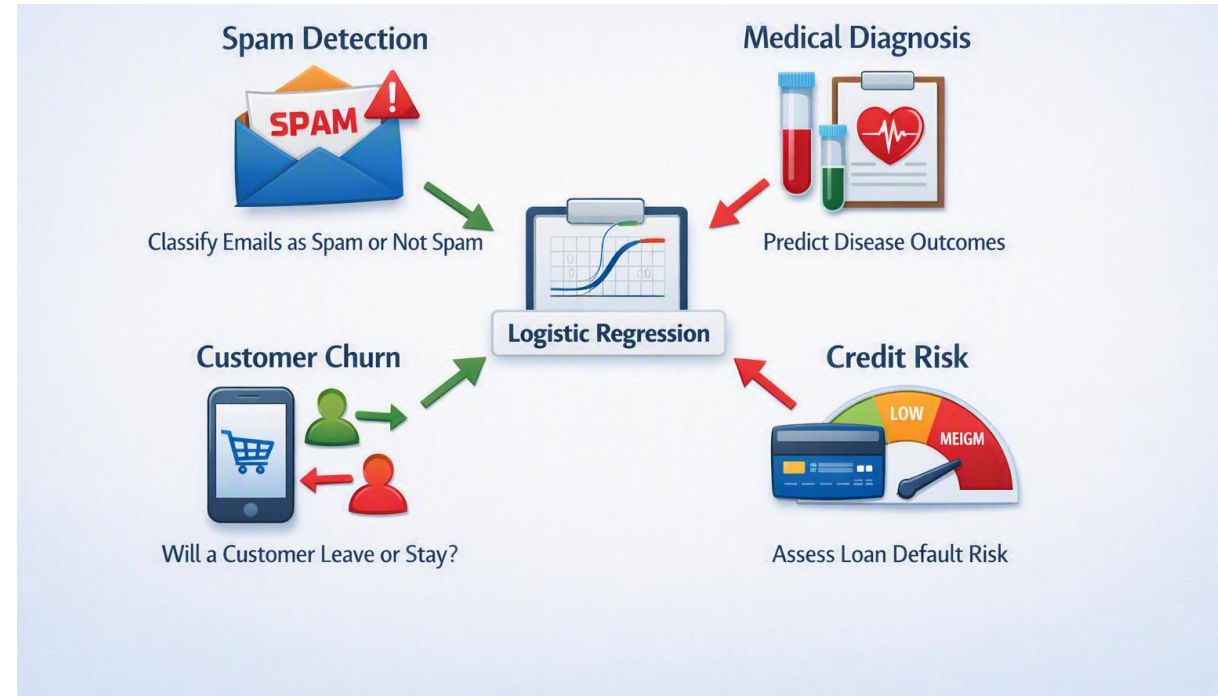
- ▶ Logistic Regression is useful when the target variable represents a binary outcome and we need an interpretable probability-based prediction.

Why Use

- Binary Classification
- Probability Output
- Easy to Interpret
- Fast & Efficient

Real-World Applications

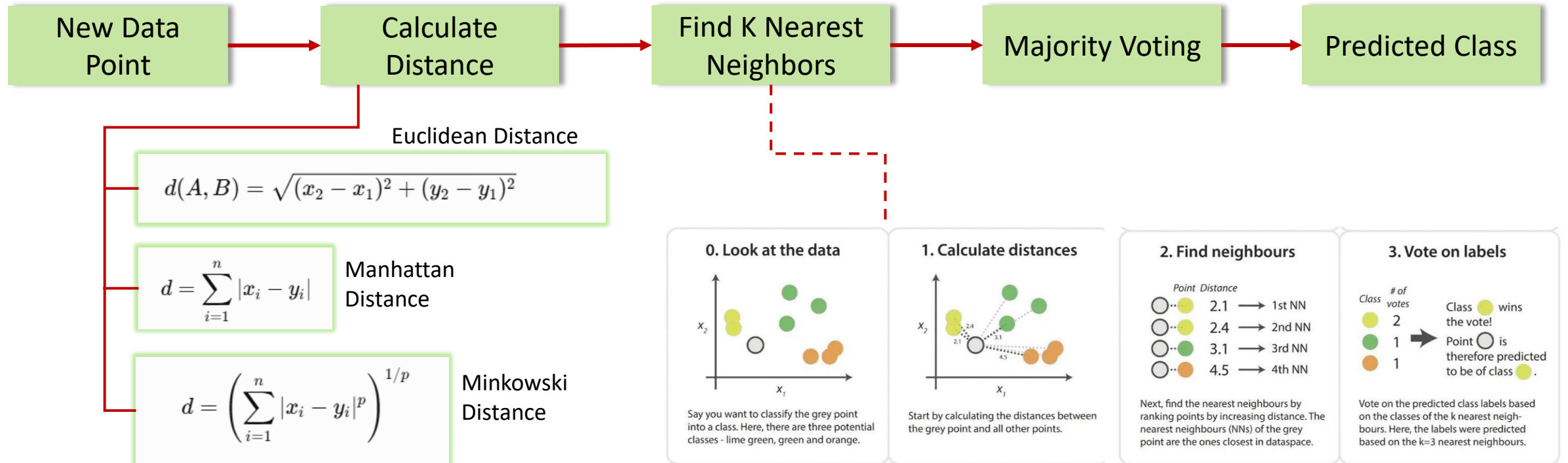
- Medical Diagnosis
- Spam Detection
- Credit Risk
- Customer Churn



Logistic Regression is a strong baseline classifier because it is simple, fast, interpretable, and produces useful probability estimates

k-Nearest Neighbors (kNN)

- kNN is a distance-based classification algorithm that assigns a class to a new data point based on the classes of its nearest neighbors.



k-Nearest Neighbors (kNN)

What is K?

K = Number of nearest neighbors considered when making a prediction

K = 1

- Very sensitive to individual points
- Can **overfit**
- Low bias, high variance

Large K

- Smoother decision boundary
- Can **underfit**
- Higher bias, lower variance

Try different K values

Evaluate on validation data

Best K

Compare performance

Lazy Learning in kNN

Means that kNN does not learn a model during the training phase. Instead, it stores the training data and waits until a new observation needs to be classified.

Strengths

- Simple and intuitive
- No explicit training phase
- Can model nonlinear decision boundaries
- Works well for smaller datasets

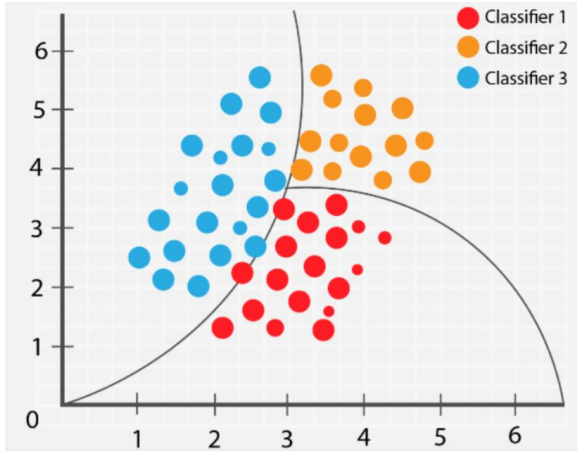
Limitations

- Prediction can be **slow**
- Sensitive to **feature scaling**
- Sensitive to irrelevant features
- Large datasets require more memory
- Choosing K affects performance

kNN is simple and flexible, but its performance depends strongly on distance, feature scaling, and the choice of K

Naïve Bayes

- ▶ Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem, with the assumption that features are conditionally independent given the class.



Bayes' Theorem

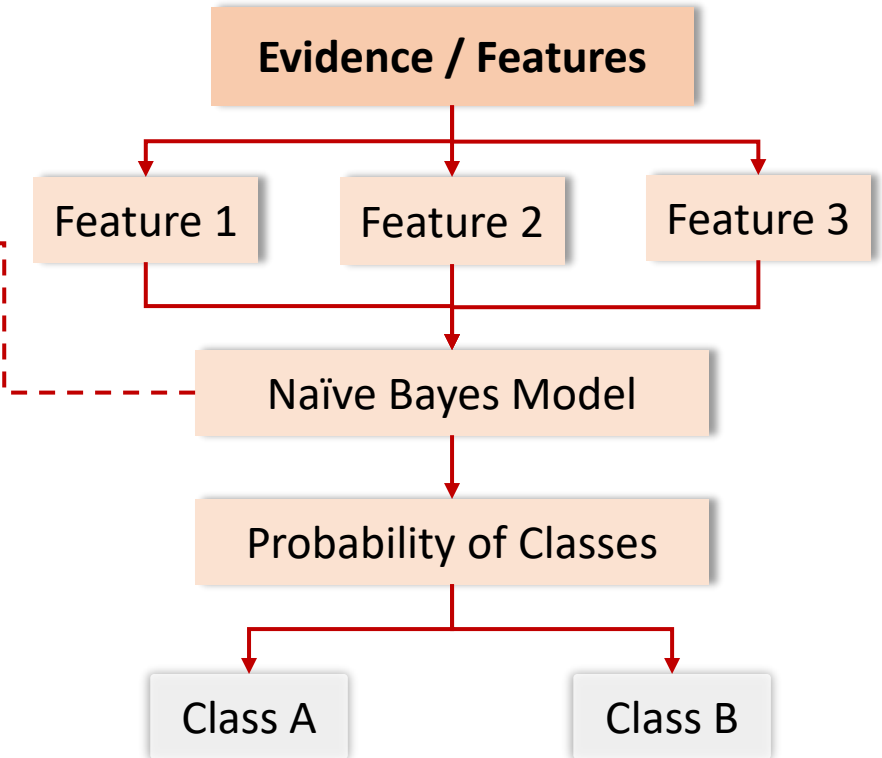
$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

$P(C|X)$ → Probability of class given the features

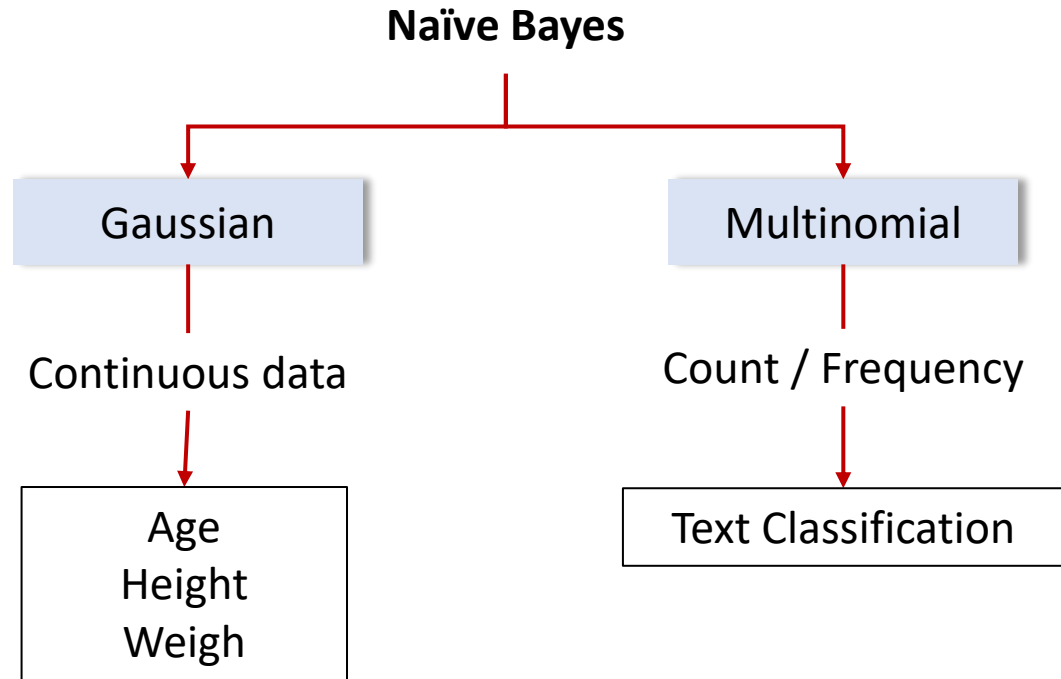
$P(X|C)$ → Probability of features given the class

$P(C)$ → Prior probability of the class

$P(X)$ → Features



Naïve Bayes



The choice of Naïve Bayes variant depends mainly on the type of features in the dataset.

Advantages

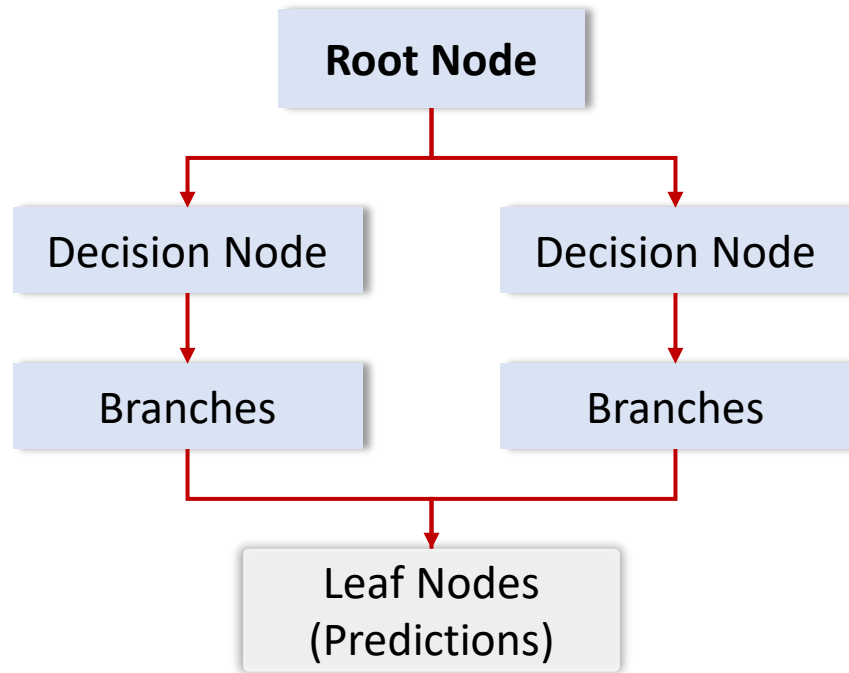
- Simple and fast
- Works well with high-dimensional data
- Requires relatively little training data
- Useful for text classification

Limitations

- Independence assumption may be unrealistic
- Correlated features can reduce effectiveness
- Probability estimates may not always be well calibrated

Decision Trees

- ▶ A Decision Tree is a supervised learning algorithm that makes predictions by following a sequence of feature-based decisions from the root to a final class.



Key Components

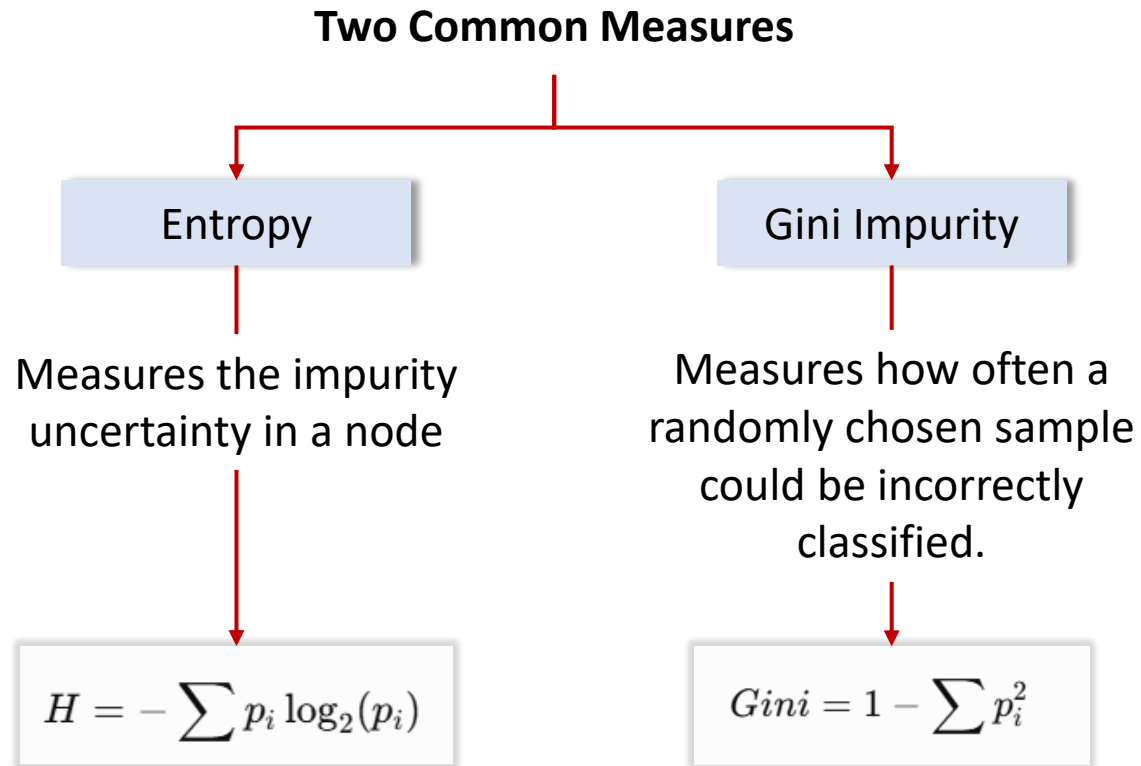
- Root Node : First and most important decision
- Decision Node : Splits data based on a feature
- Branch : Outcome of a decision
- Leaf Node : Final prediction / class

Importance

Decision Trees transform complex classification decisions into a sequence of simple, interpretable rules.

Decision Trees

- ▶ A Decision Tree selects the split that produces the purest groups of class labels.



Advantages

- Easy to understand
- Easy to visualize
- Handles nonlinear relationships
- Little preprocessing required

Limitations

- Can **overfit**
- Small data changes can produce a different tree
- Deep trees can become complex

Support Vector Machine

- ▶ SVM is a classification algorithm that finds a decision boundary that maximizes the margin between classes.

**Decision
Boundary**

$$\mathbf{w}^T \mathbf{x} + b = 0$$

$\mathbf{w} \rightarrow$ weight vector
 $\mathbf{x} \rightarrow$ feature vector
 $b \rightarrow$ bias/intercept

**Margin
Boundaries**

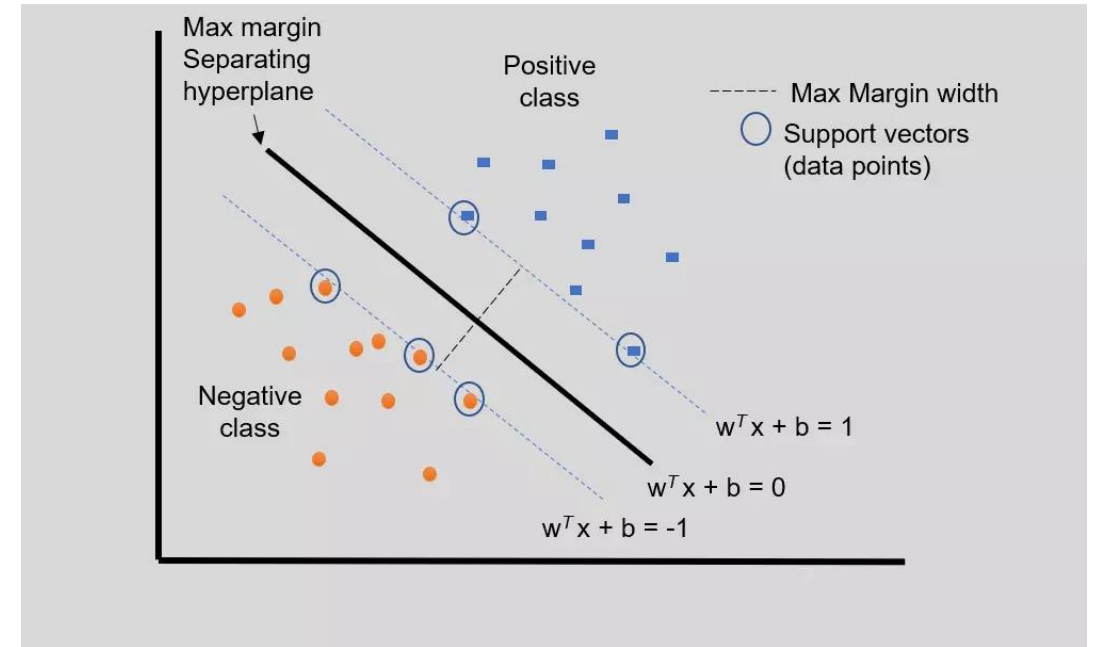
$$\mathbf{w}^T \mathbf{x} + b = +1$$

$$\mathbf{w}^T \mathbf{x} + b = -1$$

$$\text{Margin} = \frac{2}{\|\mathbf{w}\|}$$

**Distance from
a Point to the
Hyperplane**

$$d = \frac{|\mathbf{w}^T \mathbf{x} + b|}{\|\mathbf{w}\|}$$

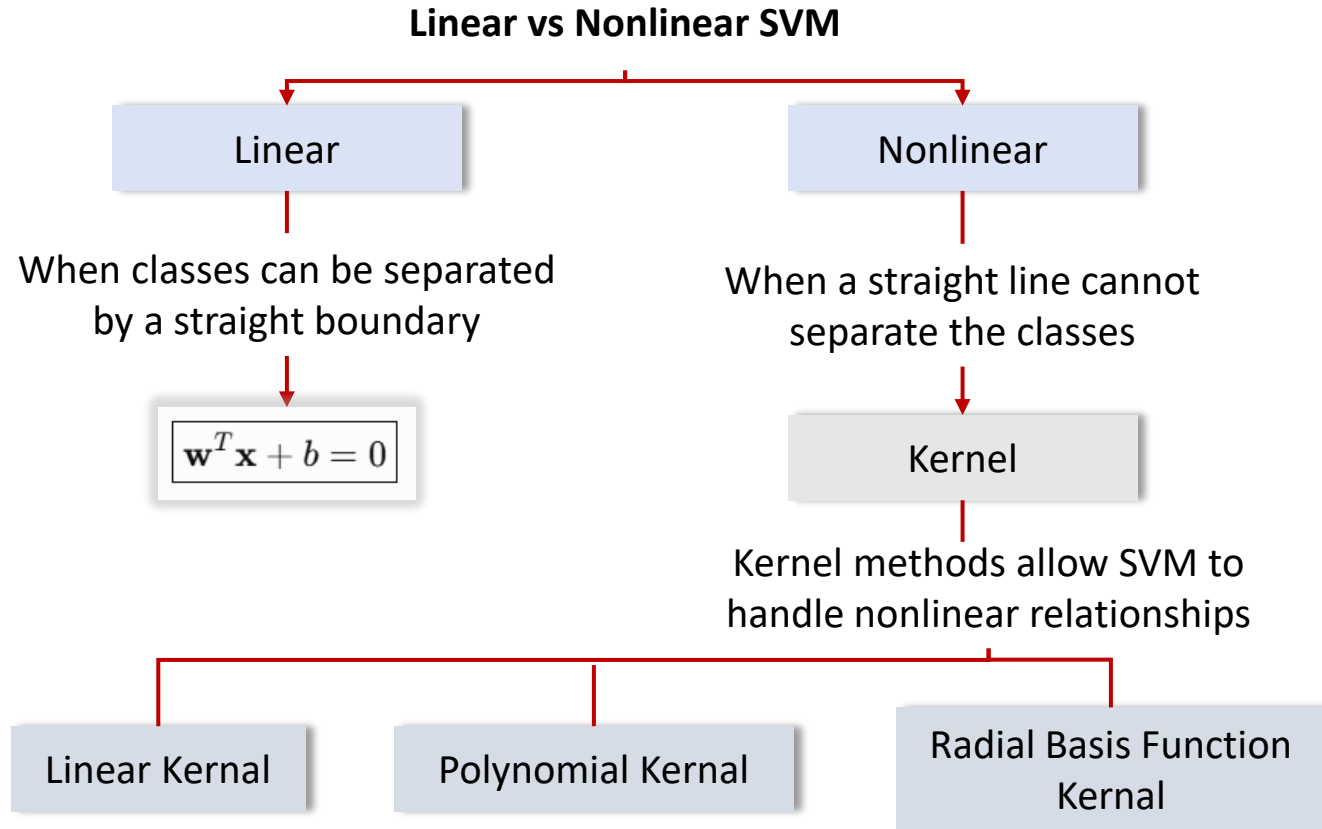


Key Terms

- **Hyperplane** : Decision boundary separating classes
- **Margin** : Distance between the boundary and closest samples.
- **Support Vectors** : Closest samples that defines the margin

Support Vector Machine

- ▶ Linear SVM - When classes can be separated by a straight boundary



Works well when

- Classes have a clear separation
- Data has many features
- A strong decision boundary is required
- Dataset is small-to-medium sized
- Features can be appropriately scaled

Less suitable when

- Dataset is extremely large
- Data contains substantial noise/overlap
- Features are not properly scaled

Comparing Classification Algorithms

- ▶ No single classification algorithm is best for every dataset; the choice depends on the data, decision boundary, interpretability, and computational requirements.

Algorithm	Task	Decision Boundary	Interpretability	Training Speed	Prediction Speed	Scalability
Linear Regression	Regression	Linear	Very High	Fast	Fast	High
Logistic Regression	Classification	Linear	High	Fast	Fast	High
kNN	Classification	Nonlinear	Medium	Very Fast	Slow	Low
Naïve Bayes	Classification	Usually simple	Medium	Very Fast	Very Fast	High
Decision Tree	Classification	Nonlinear	Very High	Fast	Fast	Medium
SVM	Classification	Linear / Nonlinear	Low–Medium	Moderate	Fast	Medium

Classification Performance Evaluation

- Classification metrics measure how well a classifier's predictions match the actual class labels.

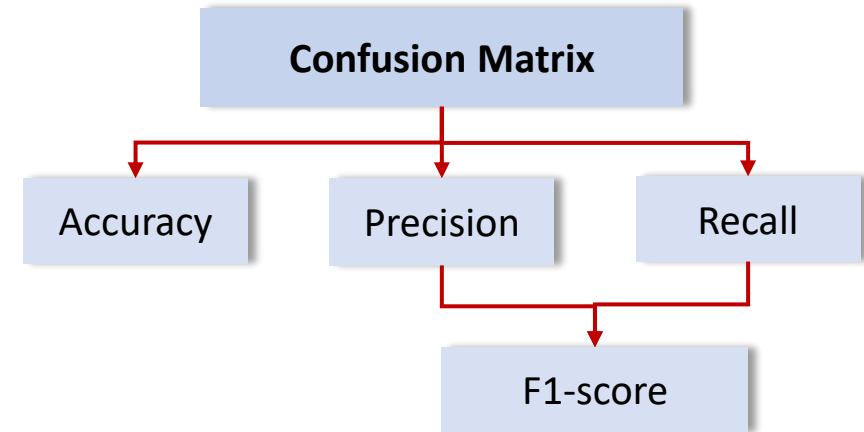
Confusion matrix compares
Actual Class vs Predicted Class

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

True Positive (TP)
True Negative (TN)
False Positive (FP)
False Negative (FN)

		Predicted	
		Disease	Healthy
Actual	Disease	TP 40	FN 10
	Healthy	FP 5	TN 45

TP (40)
Patient has the disease → Model predicts Disease
FN (10)
Patient has the disease → Model predicts Healthy
FP (5)
Patient is healthy → Model predicts Disease
TN(45)
Patient is healthy → Model predicts Healthy



Classification Performance Evaluation

Accuracy

Overall correctness

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Of the predicted positives, how many were actually positive

$$Precision = \frac{TP}{TP + FP}$$

Recall

Of all actual positives, how many did the model detect

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

Balances Precision and Recall

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

Metric	Main Question
Accuracy	How many predictions are correct?
Precision	When model says Positive, how often is it right?
Recall	How many actual Positives did it find?
F1	How well are Precision & Recall balanced?

ROC Curve & AUC

ROC Curve shows how the classifier's performance changes as the decision threshold changes

Axes

X-axis → False Positive Rate (FPR)

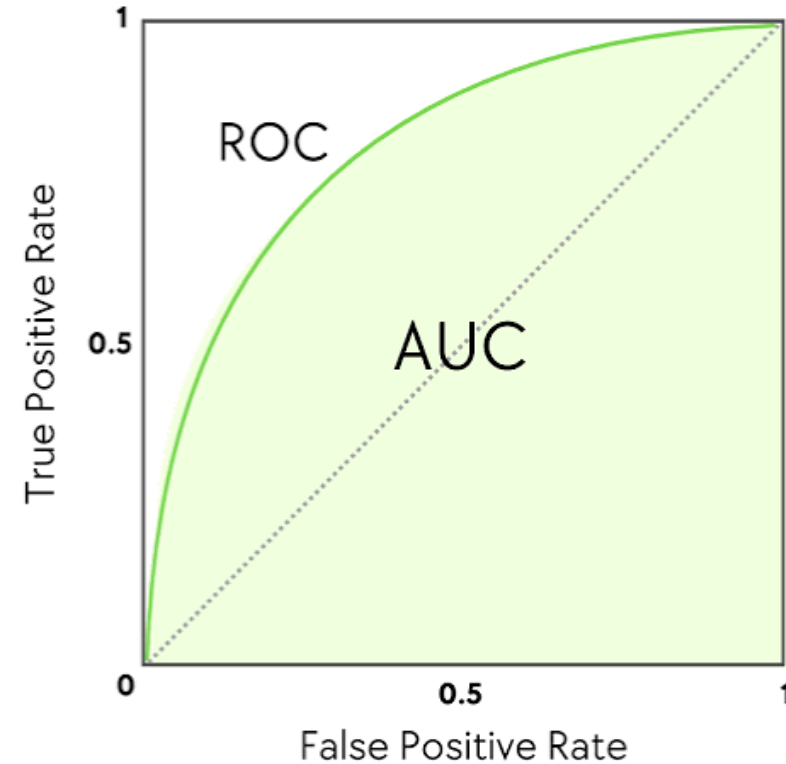
$$FPR = \frac{FP}{FP + TN}$$

Y-axis → True Positive Rate (TPR / Recall)

$$TPR = \frac{TP}{TP + FN}$$

AUC Area Under the ROC Curve

AUC	Interpretation
1.0	Perfect separation
0.9	Excellent
0.7	Moderate
0.5	Random
< 0.5	Worse than random



Precision–Recall (PR) Curve

- ▶ A PR curve shows the trade-off between Precision and Recall across different classification thresholds.

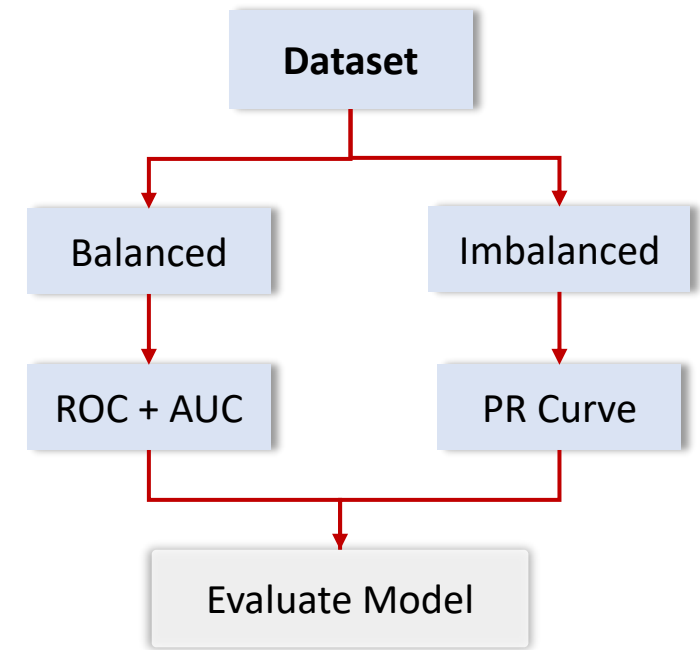
Axes

X-axis → Recall

$$Recall = \frac{TP}{TP + FN}$$

Y-axis → Precision

$$Precision = \frac{TP}{TP + FP}$$



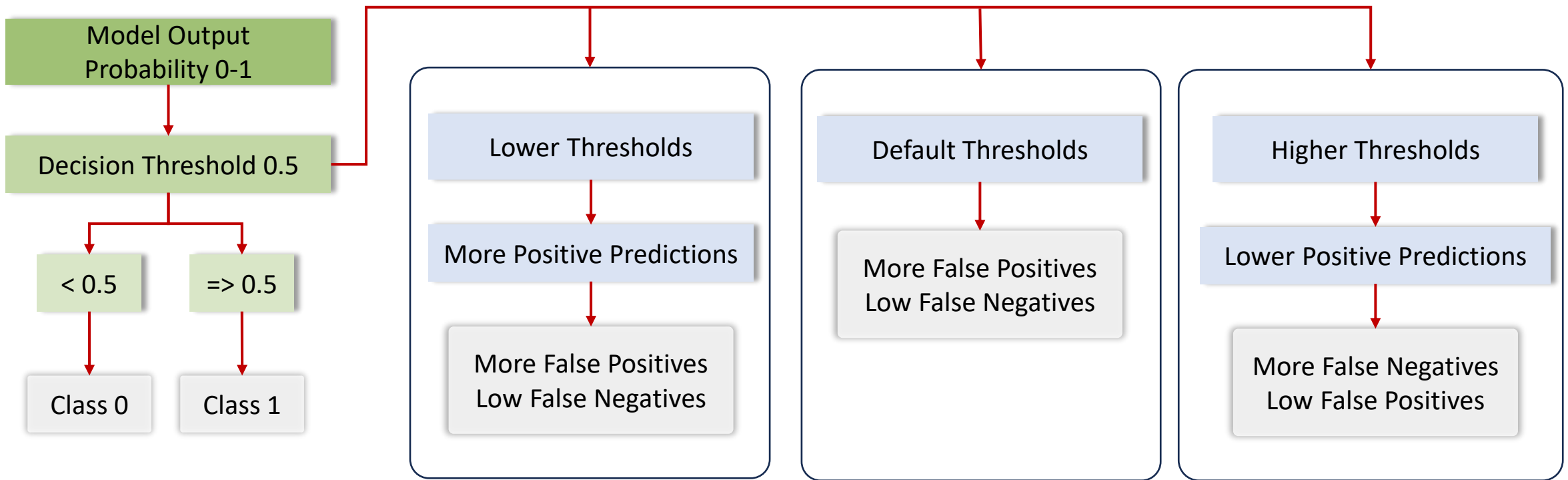
	ROC Curve	PR Curve
Y-axis	TPR / Recall	Precision
X-axis	FPR	Recall
Focus	Overall class separation	Positive-class performance
Useful for	Balanced datasets	Imbalanced datasets

For imbalanced classification, the Precision–Recall curve helps reveal how well the model identifies the minority/positive class.

Decision Threshold

- ▶ The decision threshold determines when a predicted probability is converted into a positive or negative class.

From Probability → Class



The best threshold depends on the relative cost of false positives and false negatives in the application.

Bias–Variance

- ▶ A good model balances simplicity and flexibility so that it learns the underlying pattern without memorizing the training data.

Underfitting

Model hasn't learned enough from the data.

Characteristics

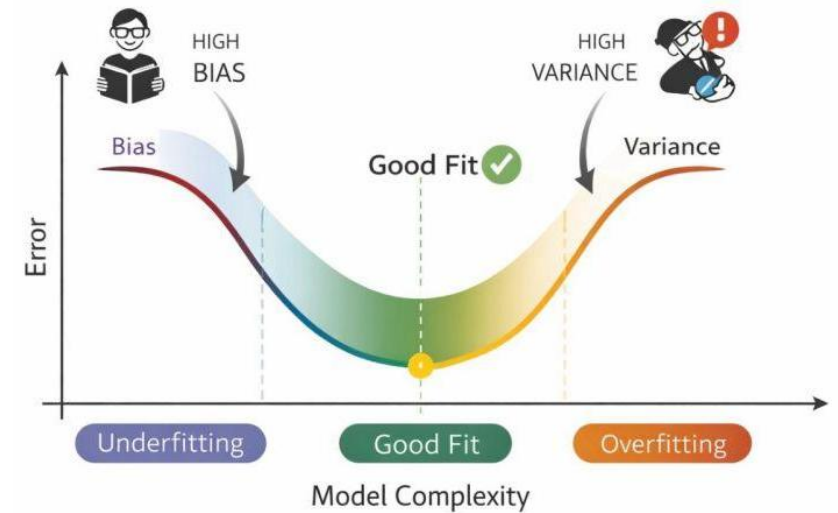
- High Bias
- Poor training performance
- Poor test performance

Good Fit

Model captures the underlying pattern

Characteristics

- Appropriate complexity
- Good generalization
- Good training + test performance



Underfitting

Model hasn't learned enough from the data.

Characteristics

- High Bias
- Poor training performance
- Poor test performance

The goal is not to make the model as complex as possible, but to achieve good performance on unseen data.

Summary

In this lecture, we learned:

- ▶ The supervised learning workflow and how classification differs from regression
- ▶ What classification is, including binary, multiclass, and multilabel problems
- ▶ Why linear regression is unsuitable for classification tasks
- ▶ How **Logistic Regression, kNN, Naïve Bayes, Decision Trees, and SVM** approach classification
- ▶ How to compare classifiers by interpretability, speed, scalability, and decision boundary
- ▶ How to evaluate classifiers using confusion matrix, accuracy, precision, recall, and F1-score
- ▶ How to interpret ROC, AUC, and Precision-Recall curves, especially for imbalanced data
- ▶ How decision threshold choice and the bias-variance trade-off affect model behavior

Thank You